

**STORAGE OF PERISHABLE FOOD USING AN IMPROVISED MODIFIED
ATMOSPHERIC STORAGE METHOD**

BY

IGBINNOSA JOEL OSAHERHUMWEN

2016/1/62028EA

**DEPARTMENT OF AGRICULTURAL AND BIO RESOURCES ENGINEERING,
SCHOOL OF INFRASTRUCTURE PROCESSING AND ENGINEERING
TECHNOLOGY, FEDERAL UNIVERSITY OF TECHNOLOGY, MINNA**

MAY 2023

**STORAGE OF PERISHABLE FOOD USING AN IMPROVISED MODIFIED
ATMOSPHERIC STORAGE METHOD**

BY

IGBINNOSA JOEL OSAHERHUMWEN

2016/1/62028EA

**DEPARTMENT OF AGRICULTURAL AND BIO RESOURCES ENGINEERING,
SCHOOL OF INFRASTRUCTURE PROCESSING AND ENGINEERING
TECHNOLOGY, FEDERAL UNIVERSITY OF TECHNOLOGY, MINNA**

MAY 2023

DECLARATION

I, **IGBINNOSA JOEL OSAHERHUMWEN** hereby declare that this project work was absolutely conducted by me, under the supervision of PROF P. A. IDAH at the department of Agricultural and Bioresources Engineering, Federal University of Technology Minna, Niger State. This work has not been presented wholly or impact for the award of any degree elsewhere. Information and data used in this project was derived from published work of other researchers.

IGBINNOSA JOEL OSAHERHUMWEN

2016/1/62028EA

SIGN & DATE

CERTIFICATION

This is to certify that the project titled “**STORAGE OF PERISHABLE FOOD USING AN IMPROVISED MODIFIED ATMOSPHERIC STORAGE METHOD**” was fully carried out by IGBINNOSA JOEL OSAHERHUMWEN with matriculation number 2016/1/62028EA under the close supervision of ENGR PROF P. A. IDAH

and submitted to the Agricultural and Bio Resources Engineering Department of Federal University of Technology Minna in partial fulfilment of the requirement for the award of Bachelor of Engineering (B. Eng) degree in Agricultural and Bio Resources Engineering.

Engr. Prof. P. A. Idah
PROJECT SUPERVISOR

Date

Engr. Dr. S. M. Dauda
HEAD OF DEPARTMENT

Date

EXTERNAL EXAMINER

Date

DEDICATION

This project is dedicated to my parents, MR and MRS IGBINNOSA and every single person that has supported me in one way or the other in this journey.

ACKNOWLEDGEMENTS

My sincere gratitude goes to my supervisor, Engr. Prof. P. A. Idah for his time, effort and guidance in this project. His advice, criticism and corrects helps me to fine tune this project to the standard it is today. My thanks also goes to the staff of the Department of Agricultural and Bio-resources Engineering, School of Infrastructure Process Engineering Technology, Federal University of Technology Minna, Niger State for providing the environment to carryout this research inquiry. A special mention goes to Mr Musa Bulus Baba from the WAFT department for his guidance throughout the project. I would also like to thank my parents for their support and encouragement throughout my studies in this school.

Finally, and most importantly, I give all glory, honour, and thanks to the Most High God for giving me the will and strength to see this project and implicationally this degree to the end. Thank you, FATHER!

ABSTRACT

Current storage methods in rural and semi-rural environments are inefficient and prone to losses especially when storing perishable foods. This is because of the high cost of highly efficient storage methods. This study was carried out to evaluate an improvisation of one of such storage methods: Modified atmospheric storage. This study involved the planned storage of a pre-selected perishable product (tomato) in two groups, one acting as control and the other containing the modified atmosphere. Parameters, mostly related with acidity was then evaluated for differences in change between groups. Most parameters were in line with literature with PH values ranging from 3.91 to 4.85 and titratable acidity ranging from 0.6409 g/kg to 8.3815 g/kg. However, no significant difference was observed between different parameter groups which would have indicated better storage outcomes leading to the conclusion that such improvised modified storage method was ineffective in storing the prescribed perishable product at the stated condition of room temperature.

TABLE OF CONTENT

Cover page	i
Title pagess	ii
Declaration	iii
Certification	iv
Dedication	v
Acknowledgements	vi
Abstract	vii
Table of content	viii

CHAPTER ONE

1.0	Introduction	1
1.1	Background of study	1
1.2	Problem Statement	2
1.3	Aim and Objectives	3
1.4	Justification	3
1.5	Scope of study	3

CHAPTER TWO

2.0	Literature Review	4
2.1	Fruit and vegetables: a brief overview	4
2.2	Tomato as a case study	4
2.3	Fruit and vegetable production and uses	5
2.4	Food spoilage and losses	6
2.4.1	Microbial spoilage	6
2.4.2	Chemical spoilage	12
2.4.3	Physical spoilage	13
2.5	Food storage	13
2.5.1	History of storage methods	14
2.5.2	Food storage methods	16

CHAPTER THREE

3.0	Materials and methods	22
3.1	Materials	22
3.1.1	Apparatus	22
3.1.2	Reagents	22
3.2	Methods	22
3.2.1	Sample collection and preparation	22
3.2.2	Container preparation	23
3.2.3	Improvised atmospheric modification used	23
3.2.4	Parameters determined	23
3.2.5	Procedure for obtaining parameters.	23
3.2.5.1	Total alkalinity determination	24
3.2.5.2	Total solids determination	24
3.2.5.3	Titrateable acid determination	24
3.2.5.4	PH value determination	24

CHAPTER FOUR

4.0	Results and discussion	26
4.1	Results	26
4.2	Discussion	29

CHAPTER FIVE

5.0	Conclusion and recommendations	31
5.1	Conclusion	31
5.2	Recommendations	31

REFERENCES